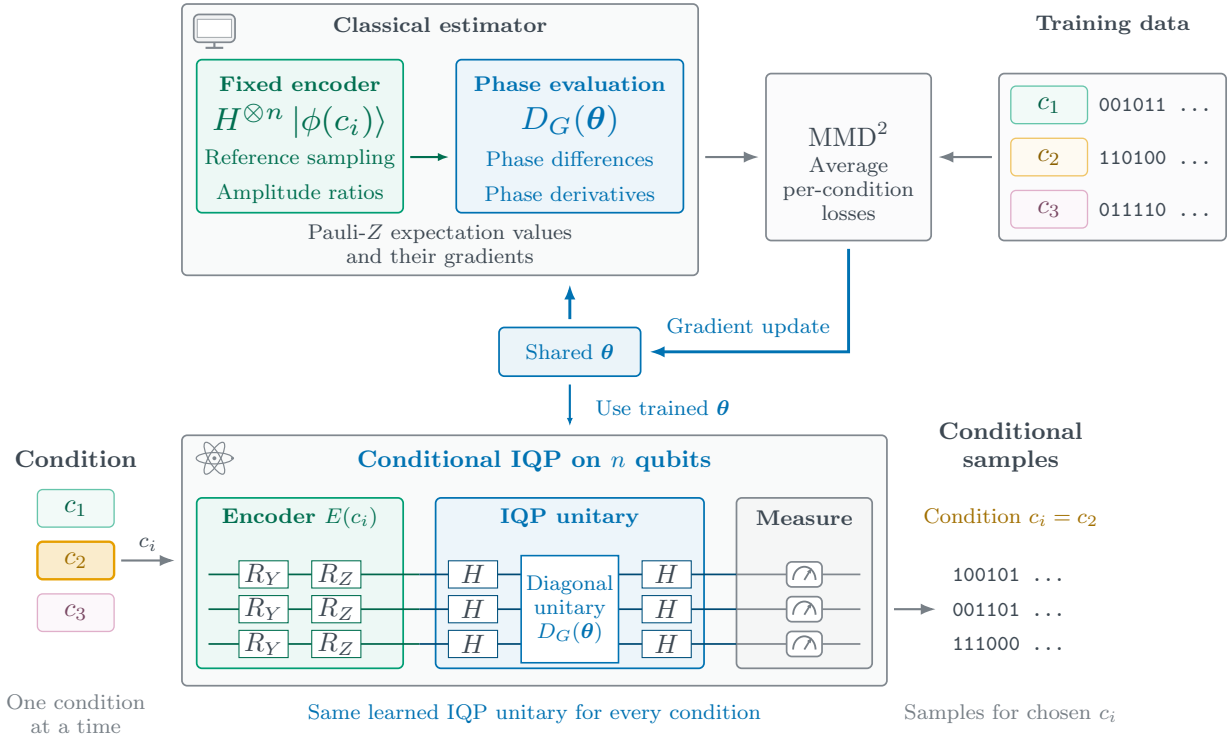

Classical estimator
Fixed encoder
$H^{\otimes n}\,|\phi(c_i)\rangle$
Reference sampling
Amplitude ratios
Phase evaluation
$D_G(\boldsymbol{\theta})$
Phase differences
Phase derivatives
Pauli-Z expectation values
and their gradients
Training data
c_1 001011 ...
c_2 110100 ...
c_3 011110 ...
MMD^2
Average
per-condition
losses
Shared $\boldsymbol{\theta}$
Gradient update
Use trained $\boldsymbol{\theta}$
Condition
c_1
c_2
c_3
c_i
One condition
at a time
Conditional IQP on n qubits
Encoder $E(c_i)$
R_Y R_Z
IQP unitary
H
Diagonal
unitary
$D_G(\boldsymbol{\theta})$
Measure
Same learned IQP unitary for every condition
Conditional
samples
Condition $c_i = c_2$
100101 ...
001101 ...
111000 ...
Samples for chosen c_i